



Software Architecture

SE205.3

Coursework
2026

Submission Deadline: 24th July 2026

Coursework Type: Group Assignment

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Coursework

This coursework for SE205.3 is worth a maximum of 40% of the module marks and contains two parts: a prototype and a report. The prototype is basically a working software and the report should contain screenshots, testing results, discussion, and most importantly individual contributions.

1. Scenario

AI-Powered Recruitment and Talent Management Platform

You have been commissioned to develop an **AI-Powered Recruitment and Talent Management Platform** for a multinational human resource consulting company. The platform aims to modernize the recruitment lifecycle by automating candidate screening, job matching, interview scheduling, talent analytics, and workforce management through a web-based system.

The platform should support recruiters, hiring managers, job seekers, and administrators while utilizing AI-driven insights to improve hiring efficiency and decision-making.

Components and Features

1. Client Application (Recruitment Platform)

Candidate Portal

- User registration and authentication.
- Professional profile creation and management.
- CV/Resume upload and management.
- Job search and application submission.
- AI-powered job recommendations based on skills and experience.
- Application tracking dashboard.

Recruiter Portal

- Job posting creation and management.
- Candidate search and filtering.
- Application review and shortlisting.
- AI-powered candidate ranking and screening.
- Interview scheduling and management.
- Communication with applicants.

Hiring Manager Dashboard

- Review shortlisted candidates.
- Interview feedback management.
- Candidate evaluation and scoring.
- Hiring decision management.

Administration Portal

- User management.
- Role and permission management.
- System monitoring.
- Recruitment analytics dashboard.
- Organization and department management.

2. Backend Services (C# ASP.NET Web API)

Database Management

Design and implement a relational database (e.g., SQL Server) to manage:

- Candidate profiles
- Job postings
- Applications
- Recruiter information
- Interview schedules
- Skill assessments
- Recruitment analytics
- Organization records

APIs for Recruitment Operations

Develop RESTful APIs for:

- User authentication and authorization
- Candidate profile management
- Resume management
- Job management
- Application processing
- Interview scheduling
- Candidate evaluation
- Reporting and analytics

Security and Access Control

Implement:

- JWT Authentication
- Role-Based Access Control (RBAC)
- Password hashing and encryption
- HTTPS communication
- Audit logging
- Data privacy compliance measures

AI and Analytics Services

Implement APIs for:

- Resume parsing
- Skill extraction
- Candidate-job matching
- Candidate ranking
- Recruitment performance analytics
- Hiring trend analysis

3. Integration with External Services

AI Service Integration

Integrate AI services for:

- Resume analysis
- Candidate scoring
- Skill extraction
- Job recommendation generation
- Automated feedback generation

Communication Services

Integrate:

- Email notifications
- SMS notifications
- Interview reminders
- Application status updates

Calendar Integration

Integrate with:

- Microsoft Outlook Calendar
- Google Calendar

For interview scheduling and availability management.

Cloud Storage Integration

Provide secure storage for:

- Candidate resumes
- Certifications
- Supporting documents

Responsive Design

Ensure the frontend application:

- Supports desktop, tablet, and mobile devices.
- Provides accessibility support.
- Delivers a consistent user experience.

Tasks for Development

Task 1: Client Application Development

Design and develop frontend components using HTML5, CSS3, and JavaScript frameworks such as Angular, React, or Vue.

The client application should:

- Provide responsive user interfaces.
- Implement client-side validation.
- Support secure authentication workflows.
- Deliver intuitive user experiences for all user roles.
- Include proper error handling and user feedback mechanisms.

Conduct usability testing and improve the user experience based on collected feedback.

Task 2: Backend API Development

Design and implement RESTful APIs using C# ASP.NET Web API.

The backend should:

- Manage recruitment workflows.
- Handle candidate applications.
- Support authentication and authorization.
- Implement AI-related service integrations.
- Maintain system reliability through logging and monitoring.

Ensure APIs are documented using tools such as Swagger/OpenAPI.

Task 3: Integration and Testing

Integrate frontend applications with backend services to ensure seamless functionality.

Conduct:

Unit Testing

- API endpoint testing
- Business logic validation
- Authentication testing

Integration Testing

- Frontend-backend communication
- Database operations
- Third-party service integrations

User Acceptance Testing (UAT)

Validate end-to-end recruitment workflows including:

- Job application submission
- Candidate screening
- Interview scheduling
- Recruitment reporting

2. Deliverables

I. Prototype

The prototype should be a fully functional solution containing:

Client Application

- Candidate Portal

- Recruiter Portal
- Hiring Manager Dashboard
- Administrator Dashboard

Backend Services

- ASP.NET Web API
- SQL Server Database

API Testing

The backend services must be tested using third-party tools such as:

- Postman
- Swagger

Evidence of testing should be included within the report.

II. Report

The report should contain the following sections.

Architectural Design

The following diagrams are mandatory:

1. Use Case Diagram
2. Class Diagram (Mandatory)
3. Deployment Diagram

Design Patterns

Discuss the design patterns used within the system, such as:

- Repository Pattern
- Unit of Work Pattern
- Dependency Injection
- Factory Pattern
- Strategy Pattern

Architectural Decisions

Discuss:

- System architecture selection
- Database design decisions
- Security mechanisms

- API design principles
- AI integration approach
- Scalability considerations
- Assumptions and constraints

System Demonstration

Include screenshots of:

- User interfaces
- API testing results
- Authentication workflows
- Database implementation
- AI-related functionalities

Individual Contributions

Each group member must clearly explain:

- Responsibilities undertaken
- Features implemented
- Testing contributions
- Challenges encountered
- Lessons learned

The report should contain standard references, figures, tables, and diagrams where appropriate.

3. Assessment Criteria

Category	Criteria	LOs	Marks
Prototype (Group)	• Quality of the program (coding styles, OOP concepts, testing results, etc.) • Executable program (service and client)	LO3	45%
Report (Group)	• Architectural diagrams • Design patterns • Discussion on the architectural decisions	LO1 LO2	45%
Contribution (Individual)	• Individual contribution	LO2	10%

4. Submission Type

The coursework should be uploaded to the NLEARN as a single zip file submission containing the Prototype, Prototype Demonstration Video Link and the Report:

- Upload the Prototype(Code) as a zip file
- Upload your video recording to your NSBM OneDrive. Please note that you should provide access to the video link for all users (evaluators) or you will get zero marks.
- Get the OneDrive link and copy and paste it to the Word document and upload the Word document.
- Upload the Final Report as a PDF file

Bonus Features (Optional)

Teams may implement additional advanced features such as:

- AI-powered interview question generation
- Candidate sentiment analysis
- AI chatbot for candidate support
- Predictive hiring analytics
- Skill-gap analysis and learning recommendations
- Recruitment KPI dashboards
- Multi-company recruitment support
- Video interview analysis

These features are optional but may strengthen the quality, innovation, and architectural complexity of the